

Project 07: RAG + MCP — Complete Command

Reference

Every `gcloud`, `gsutil`, `terraform`, `psql`, `git`, and related terminal command used across this project, in roughly chronological order, grouped by purpose.

1. Terraform — Init, Plan, Apply, State Management

```
EO
SE

bash
terraform init
terraform plan
terraform apply
terraform plan -target="google_artifact_registry_repository.mcp_server_repo"
terraform apply -target="google_artifact_registry_repository.mcp_server_repo"
```

State recovery (after a mid-apply network drop)

```
EO
SE

bash
terraform state push errored.tfstate
terraform force-unlock <LOCK_ID>
terraform import google_sql_database_instance.rag_pg rag-pipe-mcp/rag-pipe-mcp-pg
```

Cloud Shell terraform install (fresh session)

```
EO
SE

bash
wget -O - https://apt.releases.hashicorp.com/gpg | sudo gpg --dearmor -o /usr/share/keyrings/hashicorp-archive-keyring.gpg
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] https://apt.releases.hashicorp.com terraform mainline" | sudo tee /etc/apt/sources.list.d/hashicorp.list
sudo apt update && sudo apt install terraform
```

Shell cache issue fix (terraform binary conflict)

```
EO
SE

bash
which terraform
dpkg -L terraform | grep bin
echo $PATH
type -a terraform
hash -r
```

2. GCP API Enablement

```
EO
SE

bash
gcloud services enable cloudresourcemanager.googleapis.com
gcloud services list --enabled --filter="name:cloudresourcemanager.googleapis.com"
```

3. IAM — Role Bindings (Project-Level)



bash

```
gcloud projects add-iam-policy-binding rag-pipe-mcp --member="serviceAccount:514728048358-compute@developer.gcpcloud.com"
```

```
gcloud projects add-iam-policy-binding rag-pipe-mcp --member="serviceAccount:514728048358-compute@developer.gcpcloud.com"
```

```
gcloud projects add-iam-policy-binding rag-pipe-mcp --member="serviceAccount:514728048358-compute@developer.gcpcloud.com"
```

```
gcloud projects add-iam-policy-binding rag-pipe-mcp --member="serviceAccount:rag-pipe-mcp-pipeline@rag-pipe-mcp.iam.gserviceaccount.com"
```

```
gcloud projects add-iam-policy-binding rag-pipe-mcp --member="serviceAccount:rag-pipe-mcp-pipeline@rag-pipe-mcp.iam.gserviceaccount.com"
```

```
gcloud projects add-iam-policy-binding rag-pipe-mcp --member="serviceAccount:rag-pipe-mcp-pipeline@rag-pipe-mcp.iam.gserviceaccount.com"
```

```
gcloud projects add-iam-policy-binding rag-pipe-mcp --member="serviceAccount:514728048358@cloudshell.gcpcloud.com"
```

Check current bindings for an identity



bash

```
gcloud projects get-iam-policy rag-pipe-mcp --flatten="bindings[].members" --filter="bindings.members:514728048358-compute@developer.gcpcloud.com"
```

```
gcloud projects get-iam-policy rag-pipe-mcp --flatten="bindings[].members" --filter="bindings.members:514728048358-compute@developer.gcpcloud.com"
```

4. IAM — Service Account-Level Bindings (Impersonation & Cross-SA Acting)



bash

```
gcloud iam service-accounts add-iam-policy-binding rag-pipe-mcp-pipeline@rag-pipe-mcp.iam.gserviceaccount.com --member="serviceAccount:rag-pipe-mcp-pipeline@rag-pipe-mcp.iam.gserviceaccount.com"
```

```
gcloud iam service-accounts add-iam-policy-binding rag-pipe-mcp-pipeline@rag-pipe-mcp.iam.gserviceaccount.com --member="serviceAccount:rag-pipe-mcp-pipeline@rag-pipe-mcp.iam.gserviceaccount.com"
```

```
gcloud iam service-accounts add-iam-policy-binding rag-pipe-mcp-pipeline@rag-pipe-mcp.iam.gserviceaccount.com --member="serviceAccount:rag-pipe-mcp-pipeline@rag-pipe-mcp.iam.gserviceaccount.com"
```

Check bindings on the SA itself



bash

```
gcloud iam service-accounts get-iam-policy rag-pipe-mcp-pipeline@rag-pipe-mcp.iam.gserviceaccount.com --flatten="bindings[].members" --filter="bindings.members:rag-pipe-mcp-pipeline@rag-pipe-mcp.iam.gserviceaccount.com"
```

Local impersonation (ADC)



bash

```
gcloud auth application-default login --impersonate-service-account=rag-pipe-mcp-pipeline@rag-pipe-mcp.iam.gserviceaccount.com
```

```
gcloud auth list
```

5. Cloud SQL



```
bash
gcloud sql instances list
gcloud sql instances describe rag-pipe-mcp-pg --format="value(state)"
gcloud sql users set-password postgres --instance=rag-pipe-mcp-pg --password=hgfdsqetyi
gcloud sql connect rag-pipe-mcp-pg --user=postgres --database=ragdb
```

Cloud SQL Auth Proxy (manual, background process)



```
bash
cloud-sql-proxy rag-pipe-mcp:europa-west4:rag-pipe-mcp-pg --port 9470 &
cloud-sql-proxy rag-pipe-mcp:europa-west4:rag-pipe-mcp-pg --port 9470 --private-ip &
```

psql direct connection (through the proxy)



```
bash
psql "host=127.0.0.1 port=9470 dbname=ragdb user=postgres sslmode=disable"
```

Process/port diagnostics (proxy troubleshooting)



```
bash
jobs
ss -tlnp | grep 9470
kill -9 <PID>
kill -9 %1
```

6. GCS / gsutil



```
bash
gsutil cp your-file.pdf gs://rag-pipe-mcp-ingestion/
gsutil cp data/*.pdf gs://rag-pipe-mcp-ingestion/
gsutil -m cp -r data/* gs://rag-pipe-mcp-ingestion/
```

7. Artifact Registry



```
bash
gcloud artifacts docker images list europa-west4-docker.pkg.dev/rag-pipe-mcp/mcp-server-repo/mcp-ser
```

8. Cloud Build



```
bash
gcloud builds submit --config cloudbuild.yaml .
gcloud builds list --limit=5 --format="table(id,status,substitutions.COMMIT_SHA)"
```

9. Cloud Run — Service (MCP Server)

```
EO
SE

bash
gcloud run deploy rag-pipe-mcp-mcp-server --image=europe-west4-docker.pkg.dev/rag-pipe-mcp/mcp-se
gcloud run services describe rag-pipe-mcp-mcp-server --region=europe-west4 --format="value(status.trac
gcloud run services describe rag-pipe-mcp-mcp-server --region=europe-west4 --format="value(status.late
gcloud run services describe rag-pipe-mcp-mcp-server --region=europe-west4 --format="value(spec.temp
gcloud run revisions describe rag-pipe-mcp-mcp-server-00006-fwd --region=europe-west4 --format="valu
gcloud run services get-iam-policy rag-pipe-mcp-mcp-server --region=europe-west4
gcloud run services logs read rag-pipe-mcp-mcp-server --region=europe-west4 --limit=50
```

10. Cloud Run — Jobs (Embed / Create Extension)

```
EO
SE

bash
gcloud run jobs execute rag-pipe-mcp-embed-job --region=europe-west4
gcloud run jobs execute rag-pipe-mcp-embed-job --region=europe-west4 --args=create_extension.py --up
gcloud run jobs executions list --job=rag-pipe-mcp-embed-job --region=europe-west4
gcloud run jobs executions describe rag-pipe-mcp-embed-job-wn574
```

11. Logging — Reading Container/Revision-Specific Logs

```
EO
SE

bash
gcloud logging read 'resource.type="cloudsql_database" AND resource.labels.database_id="rag-pipe-mcp
gcloud logging read 'resource.type="cloud_run_job" AND resource.labels.job_name="rag-pipe-mcp-embed
gcloud logging read 'resource.type="cloud_run_revision" AND resource.labels.service_name="rag-pipe-mc
```

12. Testing the Deployed Endpoint



bash

```
curl -i https://rag-pipe-mcp-mcp-server-514728048358.europe-west4.run.app/mcp
```

```
curl -i -X POST https://rag-pipe-mcp-mcp-server-514728048358.europe-west4.run.app/mcp -H "Content-Type: application/json"
```

```
curl -v -X POST https://rag-pipe-mcp-mcp-server-514728048358.europe-west4.run.app/mcp -H "Content-Type: application/json"
```

MCP Inspector



bash

```
npx @modelcontextprotocol/inspector
```

13. Git — Diagnostics for Commit/Build Mismatches



bash

```
git log --oneline -5
```

```
git show HEAD:"projects/RAG pipeline/pipeline/components/mcp_wrapper.py"
```

```
git show HEAD:"projects/RAG pipeline/pipeline/components/mcp_wrapper.py" | Select-String "middleware"
```

```
git branch --show-current
```

```
git status
```

```
git add "projects/RAG pipeline/pipeline/components/create_extension.py"
```

```
git commit -m "add pgvector extension setup script"
```

```
git push
```

14. Python Environment / Package Install



bash

```
pip install pypdf
```

```
pip install anthropic
```

```
pip install mistralai
```

```
python3 embed.py
```

```
python .\embed.py
```

```
python3 test_endpoint.py
```

```
python3 test_endpoint_by_model.py
```

15. Environment Variables (PowerShell / bash session setup)

PowerShell



powershell

```
$env:PROJECT_ID = "rag-pipe-mcp"
$env:REGION = "europe-west4"
$env:CLOUD_SQL_CONNECTION_NAME = "rag-pipe-mcp:europe-west4:rag-pipe-mcp-pg"
$env:DB_NAME = "ragdb"
$env:DB_USER = "rag-pipe-mcp-pipeline@rag-pipe-mcp.iam.gserviceaccount.com"
$env:INGESTION_BUCKET_NAME = "rag-pipe-mcp-ingestion"
$env:ANTHROPIC_API_KEY = "your-actual-api-key-here"
$env:MISTRALAI_API_KEY = "your-actual-key"
```

bash (Cloud Shell)

```

bash
export PROJECT_ID="rag-pipe-mcp"
export REGION="europe-west4"
export CLOUD_SQL_CONNECTION_NAME="rag-pipe-mcp:europe-west4:rag-pipe-mcp-pg"
export DB_NAME="ragdb"
export DB_USER="rag-pipe-mcp-pipeline@rag-pipe-mcp.iam.gserviceaccount.com"
export INGESTION_BUCKET_NAME="rag-pipe-mcp-ingestion"
export BUCKET_NAME="${PROJECT_ID}-terraform-state"
```

16. Diagnostic / Network Checks

```

bash
ping storage.googleapis.com
nslookup storage.googleapis.com
lsuf -i:9470
```